

Biomarker discovery for retinal diseases

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Abstract

This report asks where in the retina disease leaves a detectable structural signal, and what that signal corresponds to anatomically. The data are Optical Coherence Tomography thickness profiles of 351 subjects, divided into one healthy control group and four pathological classes (age-related macular degeneration, central serous retinopathy, diabetic retinopathy and macular hole), with each retina segmented into six structural layers and each layer sampled along a horizontal cross-sectional axis.

The analysis proceeds in three stages. Scalar summaries of the profiles are first engineered and used to fit conventional classifiers under a nested cross-validation protocol, which establishes a performance reference and indicates which kinds of structure carry class information. The profiles are then treated as functional data, smoothed on a common B-spline basis and examined through pointwise class means and variances, a distance-based measure of class separability and multivariate functional principal component analysis. Finally, a confirmatory experiment refits the classifiers on progressively restricted versions of the raw measurements, in order to test directly whether the localization suggested by the first two stages holds when everything else is discarded.

The discriminative signal is concentrated in the outer retina and, spatially, around the fovea. Mean thickness of the outer nuclear and photoreceptor inner region (layer 5) is elevated in all four pathologies and is therefore a shared rather than a discriminating feature, whereas the relative spatial variability of the photoreceptor outer segment to retinal pigment epithelium region (layer 6) is specific to age-related macular degeneration. The classes differ from the controls in spread as well as in level, but the effect size is modest: class membership accounts for about 17% of the total functional variability, so within-class heterogeneity remains substantial and bounds how far any classifier built on these profiles can go.

Acknowledgements

I would like to thank Dr. Lara Cavinato, who supervised this work, for her guidance throughout the project and for providing the data on which it rests. The OCT thickness profiles analysed here were collected by her and were made available for this project already segmented.

The work itself began as a group project carried out during the first year of my MSc at Politecnico di Milano. I am grateful to Adelaide Carnevale, Giulio Dalla Costa, Francesca Elefante, with whom that original implementation was developed. The analyses, extensions and revisions that constitute the present work were carried out independently thereafter.

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Chapter 1

Dataset presentation

The data analysed in this project consist of retinal thickness measurements obtained through Optical Coherence Tomography (OCT), a non-invasive imaging technique that yields cross-sectional views of the retina. For each subject the retina is segmented into six structural layers, each delimited by a pair of anatomical boundaries, as summarised in Table 1.1.

Layer	Boundary interval	Anatomical description
1	ilm to nflgcl	inner retinal surface to RNFL/GCL complex
2	nflgcl to iplinl	ganglion cell and inner plexiform complex
3	iplinl to inlopl	inner nuclear layer
4	inlopl to oplonl	outer plexiform layer
5	oplonl to isos	outer nuclear and photoreceptor inner region
6	isos to rpe	photoreceptor outer segment to retinal pigment epithelium

Table 1.1: The six retinal layers.

Within each layer, thickness is recorded at sequential location points along a horizontal cross-sectional axis. Each layer is therefore represented for every subject by a discretely sampled one-dimensional profile, recorded at 750 points along the cross-sectional axis. The axial pixel scale is identical for every patient, so that thickness values expressed in pixels are directly comparable across subjects and classes; and the retinal domain considered is common across patients. The full representation of a single subject thus comprises six such profiles, for a total of $6 \times 750 = 4500$ thickness values. This kind of data supports the analysis of morphological alterations at both the global and the localised scale.

The measurements analysed here were not acquired for this project: they were provided already segmented (see the Acknowledgements), and the profiles were traced by an automated pipeline, whose boundary naming convention is the one reported in Table 1.1. The analysis below therefore starts from the per-layer thickness tables. No device calibration metadata accompanied them, so thickness is reported in pixels rather than in micrometres.

The dataset comprises 351 subjects, partitioned into one healthy control group and four pathological classes:

- **Healthy**: control subjects with no retinal structural anomalies.
- **AMD** (Age-related Macular Degeneration): degeneration affecting the central macular region.
- **CSR** (Central Serous Retinopathy): accumulation of fluid beneath the retina.
- **DR** (Diabetic Retinopathy): a complication of diabetes affecting the retinal vasculature.

- **MH** (Macular Hole): a break in the macula that distorts central vision.

The five groups are markedly unbalanced, as summarised in Table 1.2: the healthy controls alone account for more than two fifths of the sample, while the smallest pathological group, AMD, contains 36 subjects. This imbalance is the reason the training and test split is stratified by class (Section 3.1), and the reason models are compared by macro-averaged F_1 rather than by overall accuracy, which the healthy group would otherwise dominate.

Class	Subjects	Share	Training	Test
Healthy	153	43.6%	115	38
DR	63	17.9%	48	15
MH	57	16.2%	43	14
CSR	42	12.0%	32	10
AMD	36	10.3%	27	9
Total	351	100%	265	86

Table 1.2: Class composition, overall and after the stratified 75/25 split.

The aim of this project is to assess whether the five groups exhibit distinct structural signatures in the retinal thickness profiles, and to identify which layers and spatial regions carry the most discriminative information.

Chapter 2

Feature engineering

Describing these profiles as functional data is the natural viewpoint, but such a representation is high dimensional and not directly suited to standard classifiers, so the information they carry is first synthesized with scalar features. Such summarization as a first step makes it possible to establish a baseline with conventional supervised models and keeps the number of features commensurate with the sample size.

The raw profiles are reduced to the following per-layer quantities:

- **Mean thickness** (`layer1` to `layer6`): the average thickness of each layer, which summarizes its global magnitude.
- **Standard deviation** (`sd_layer1` to `sd_layer6`): the spatial variability of the thickness along the cross-sectional axis.
- **Adjacent-layer ratios** (`ratio_1112` to `ratio_1516`): the ratio of mean thickness between consecutive layers, which encodes proportional dependencies between neighbouring structures.
- **Log coefficient of variation** (`logCV_layer1` to `logCV_layer6`): defined as $\log(\text{SD}/\text{Mean})$. Since the per-layer mean and standard deviation are strongly correlated, the standard deviation carries limited additional information; the log coefficient of variation is less dependent on the mean and more symmetrically distributed, and is therefore retained as an alternative measure of relative variability.

These quantities are assembled into four feature matrices of increasing richness: `means`, `means_sd`, `means_sd_ratios`, and `means_logCV_ratios`.

The features above summarise each layer in isolation, thus a further representation to also capture how the layers vary jointly within a subject is needed. For each patient the 6×6 correlation matrix across the six layer profiles has been computed, then its upper triangle has been vectorized into a 15-dimensional feature vector $(L_{1-2}, L_{1-3}, \dots, L_{5-6})$. This encodes the patient-specific dependency structure between layers and is represented as `corr_pat_spec`. All of the mentioned configurations are compared in the classification experiments.

Chapter 3

Baseline scalar-feature models

3.1 Approach outline

This section deals with the evaluation of a set of conventional classifiers on the scalar summaries introduced in Chapter 2.

These models provide the baseline against which any subsequent functional analysis can be judged. The scalar summaries discard the spatial structure of the profiles by construction, so the performance they support measures how much of the class signal survives a purely global description of each layer. A functional treatment is only worth its additional complexity if it improves on that reference.

Moreover, fitting these models is itself informative about the structure of the data. The feature matrices differ in what they encode, so comparing classifiers across them indicates which kinds of structure carry discriminative information and how far performance improves as features are added. The scalar stage can thus supply both the reference level and the first-hand structural insight.

Candidate models

Four supervised classifiers are trained, chosen to span distinct model families. All analyses in this report are carried out in R. The nested cross-validation described below is driven by the `nestedcv` package [4], which wraps `glmnet` for the penalized logistic regression and `caret` [3] for the other three.

- **Penalized multinomial logistic regression** : an elastic-net multinomial model fitted with `glmnet` [1], in the symmetric parameterisation, which estimates a coefficient vector for each of the five classes without designating a reference category.
- **Random forest**: fitted through `caret` with the `ranger` implementation [8], with 500 trees, Gini impurity as the splitting rule, and impurity-based variable importance retained.
- **Support vector machine with an RBF kernel** : fitted through `caret` with method `svmRadial`, using the kernel $K(x, x') = \exp(-\sigma \|x - x'\|^2)$.
- **k-nearest neighbours**: fitted through `caret` with method `knn`.

Every model is evaluated on every feature matrix. The classification script loops over the five matrices (`means`, `means_sd`, `means_sd_ratios`, `means_logCV_ratios`, `corr_pat_spec`) and, within each, over the four models. The comparison is then a full 4×5 model by feature-set grid of twenty fitted configurations. No model is tied to a single feature matrix in advance.

Model selection protocol

The data are first split at the patient level into a training and a test set, stratified by class, in a 75/25 proportion. The outer loop consists of ten class-stratified folds, generated once under a fixed seed and reused for all twenty configurations, so that models and feature sets are compared on identical partitions. For each outer fold the model is refitted on the nine remaining folds and its hyperparameters are re-selected from scratch by an inner ten-fold cross-validation restricted to them, the held-out fold serving only to produce predictions; any transformation of the features is estimated inside the loops and never on the full training set. Because no outer observation contributes to the selection of the hyperparameters used to predict it, the resulting estimate refers to the entire fitting procedure, tuning included, and is approximately unbiased for its performance on unseen subjects.

The hyperparameters tuned in the inner loop are listed in Table 3.1. Selection in the inner loop is driven by the multinomial log-loss for the three models fitted through `caret`, and by the default cross-validated deviance of `cv.glmnet` for the penalized logistic regression. The metric used inside the loop is deliberately not the one by which configurations are ranked: log-loss and deviance are proper scoring rules evaluated on the predicted probabilities themselves, and so vary more smoothly across a hyperparameter grid than a metric computed from hard class assignments. Macro-averaged F_1 is adopted as the reporting and ranking metric because it weights the five classes equally.

Model	Tuned	Held fixed
Penalized logistic regression	$\alpha \in \{0, 0.1, 0.25, 0.5, 0.75, 1\}$ λ over the internal path	multinomial family internal standardisation
Random forest	<code>mtry</code> $\in \{2, 4, 6, \max(2, \lfloor \sqrt{p} \rfloor)\}$ <code>min.node.size</code> $\in \{1, 5\}$	500 trees, Gini splitting rule
SVM with RBF kernel	$C \in \{2^{-3}, 2^{-1}, \dots, 2^7\}$ $\sigma \in \{2^{-5}, 2^{-3}, 2^{-1}, 2^1\}$	radial basis kernel
k -nearest neighbours	$k \in \{5, 7, \dots, 21\}$	Euclidean distance

Table 3.1: Here p is the number of columns of the feature matrix.

Performance is pooled across the outer folds rather than averaged over them. Each subject of the training set receives exactly one prediction, produced by the model whose training folds excluded that subject. The ten sets of out-of-fold predictions are then concatenated into a single vector of length equal to the training sample size, and one confusion matrix is computed from it. The selection metric is evaluated on this pooled confusion matrix. The same metric is additionally recomputed within each outer fold separately, and the standard deviation of these ten values is reported alongside the pooled figure as an indication of its stability across partitions.

Model selection in this section rests solely on these nested cross-validation results.

Selection metric

The five classes are imbalanced, so overall accuracy would be dominated by the larger groups and would reward a classifier that neglects the rarer pathologies. Configurations are therefore ranked by the macro-averaged F_1 score, which computes the F_1 score of each class in a one-versus-rest sense and averages the resulting values with equal weight, independently of class frequency:

$$\text{macro-}F_1 = \frac{1}{K} \sum_{k=1}^K \frac{2P_k R_k}{P_k + R_k}, \quad P_k = \frac{\text{TP}_k}{\text{TP}_k + \text{FP}_k}, \quad R_k = \frac{\text{TP}_k}{\text{TP}_k + \text{FN}_k}, \quad (3.1)$$

where P_k and R_k are the precision and the recall of class k , the counts TP_k , FP_k and FN_k are read off the pooled out-of-fold confusion matrix, and $K = 5$. A configuration scores well

under this metric only if it recovers every class, including the least populated ones, which is the behaviour of interest here. Balanced accuracy, the mean of the per-class sensitivities and specificities, is also reported as a secondary metric.

Applying this protocol to the twenty configurations yields one pooled out-of-fold macro- F_1 per model and feature matrix, together with its across-fold standard deviation. The consequent ranking is reported in Table 3.2.

Dataset	Model	Macro-F1	Macro-F1 sd	Balanced acc.	Balanced acc. sd
means_logCV_ratios	logreg	0.6528	0.1025	0.7812	0.0662
means_sd_ratios	logreg	0.6276	0.0920	0.7670	0.0570
means_logCV_ratios	rf	0.6028	0.1116	0.7553	0.0623
means_logCV_ratios	svm	0.5952	0.1017	0.7653	0.0634
means_sd_ratios	rf	0.5774	0.1125	0.7477	0.0656
means_sd	logreg	0.5684	0.1143	0.7317	0.0690
means	rf	0.5508	0.1381	0.7234	0.0699
means_sd	rf	0.5463	0.1156	0.7394	0.0668
means_sd_ratios	svm	0.5146	0.1126	0.7179	0.0746
means_logCV_ratios	knn	0.5025	0.1117	0.6900	0.0572
means_sd	svm	0.4998	0.0858	0.7142	0.0611
means_sd_ratios	knn	0.4980	0.1096	0.6876	0.0641
means_sd	knn	0.4958	0.1067	0.6890	0.0574
means	knn	0.4529	0.0858	0.6686	0.0514
means	logreg	0.4442	0.0588	0.6660	0.0343
means	svm	0.4130	0.0634	0.6629	0.0354
corr_pat_spec	rf	0.3762	0.1133	0.6208	0.0623
corr_pat_spec	svm	0.3614	0.0566	0.6195	0.0422
corr_pat_spec	logreg	0.3565	0.1675	0.6117	0.0937
corr_pat_spec	knn	0.2984	0.1599	0.5776	0.0769

Table 3.2: The twenty configurations ranked by decreasing macro-F1.

The ranking of Table 3.2 should not be read as a decisive comparison at its top. The three leading configurations are separated by about 0.025 macro- F_1 each, while the across-fold standard deviations of the same quantities are of the order of 0.09 to 0.11: the gaps are small relative to the variability of the estimates, so the ordering is indicative rather than a powered contrast, and the leading configurations are not statistically distinguishable on these ten folds. The coarser features of the table are more solid than its top: **corr_pat_spec** is clearly the weakest feature matrix, the ratio-augmented matrices are clearly the strongest, and the logistic regression and the random forest sit above the SVM and k -nearest neighbours throughout. What follows therefore retains **means_logCV_ratios** on the combination of its rank and the interpretability of its features, whose coefficients and importances admit a direct reading in terms of layer thickness, within-layer variability and adjacent-layer geometry, and not on a claim that it is significantly superior to the alternatives just below it.

3.2 Multinomial penalized logistic regression

The first model examined is the penalized multinomial logistic regression, fitted under the nested cross-validation protocol of Section 3.1. The coefficients discussed below are those of the configuration trained on the **means_logCV_ratios** feature matrix. All quantities reported here are obtained on the training data alone.

Specification

The model is fitted with **glmnet** in its symmetric multinomial parameterisation, which assigns a coefficient vector to each of the five classes rather than designating one of them as a reference:

$$P(Y = k | x) = \frac{\exp(\beta_{k0} + x^\top \beta_k)}{\sum_{m=1}^K \exp(\beta_{m0} + x^\top \beta_m)}, \quad k = 1, \dots, K, \quad K = 5. \quad (3.2)$$

The coefficients are estimated by minimising the penalized negative log-likelihood with an elastic-net penalty,

$$-\frac{1}{n} \sum_{i=1}^n \log P(y_i | x_i) + \lambda \sum_{k=1}^K \sum_{j=1}^p \left[\frac{1-\alpha}{2} \beta_{kj}^2 + \alpha |\beta_{kj}| \right], \quad (3.3)$$

so the penalty is not fixed in advance: the mixing parameter α is treated as a hyperparameter and tuned over the grid $\{0, 0.1, 0.25, 0.5, 0.75, 1\}$, whose endpoints are the pure ridge and pure lasso penalties. For the final refit on the whole training set, the pair retained for the penalized logistic regression on **means_logCV_ratios** is $\alpha = 1$ and $\lambda \approx 5.56 \times 10^{-3}$.

The selected $\alpha = 1$ is the upper endpoint of the grid, so the retained penalty is a pure lasso. This is what the construction of the feature matrix would lead one to expect: the adjacent-layer ratios are deterministic, though nonlinear, functions of the layer means, and each log coefficient of variation is built from the standard deviation and the mean of the same layer, so the columns of the design matrix are assembled from the same underlying per-layer quantities and carry heavily overlapping information. With redundancy of that kind, dropping columns outright is a more effective use of the penalty budget than the joint shrinkage of a large set of near-equivalent directions. The inner cross-validation accordingly selects the pure lasso endpoint.

Standardization of the log-odds

Two successive transformations are applied to the estimated coefficients before they are displayed and discussed.

The first concerns identifiability. The implementation of the model in **glmnet** estimates a coefficient vector for each of the five classes without designating a reference category. Taking Healthy as the comparison class recovers the quantity

$$\log \frac{P(Y = k | x)}{P(Y = \text{Healthy} | x)} = (\beta_{k0} - \beta_{\text{Healthy},0}) + x^\top (\beta_k - \beta_{\text{Healthy}}), \quad (3.4)$$

The code forms this difference column by column and discards the Healthy row afterwards since it is zero by construction.

The second concerns scale. The model is fitted with **standardize = TRUE**, so **glmnet** standardizes the predictors internally before optimising, which ensures that the penalty in Equation (3.3) treats every column equally instead of penalising a feature more heavily merely because its units are larger. The coefficients it returns however are back-transformed to the original scale, and on that scale they are not comparable across predictors. The coefficients are therefore rescaled after fitting by the standard deviation s_j of the corresponding column of the training feature matrix,

$$\tilde{\beta}_{kj} = (\beta_{kj} - \beta_{\text{Healthy},j}) \cdot s_j, \quad (3.5)$$

where s_j is computed on the untransformed training columns of **means_logCV_ratios**. Each $\tilde{\beta}_{kj}$ then reads as the change in the log-odds of class k relative to Healthy produced by an increase of one standard deviation in feature j . This is what makes the bars of Figure 3.1 directly comparable both across predictors within a class and across classes for a given predictor.

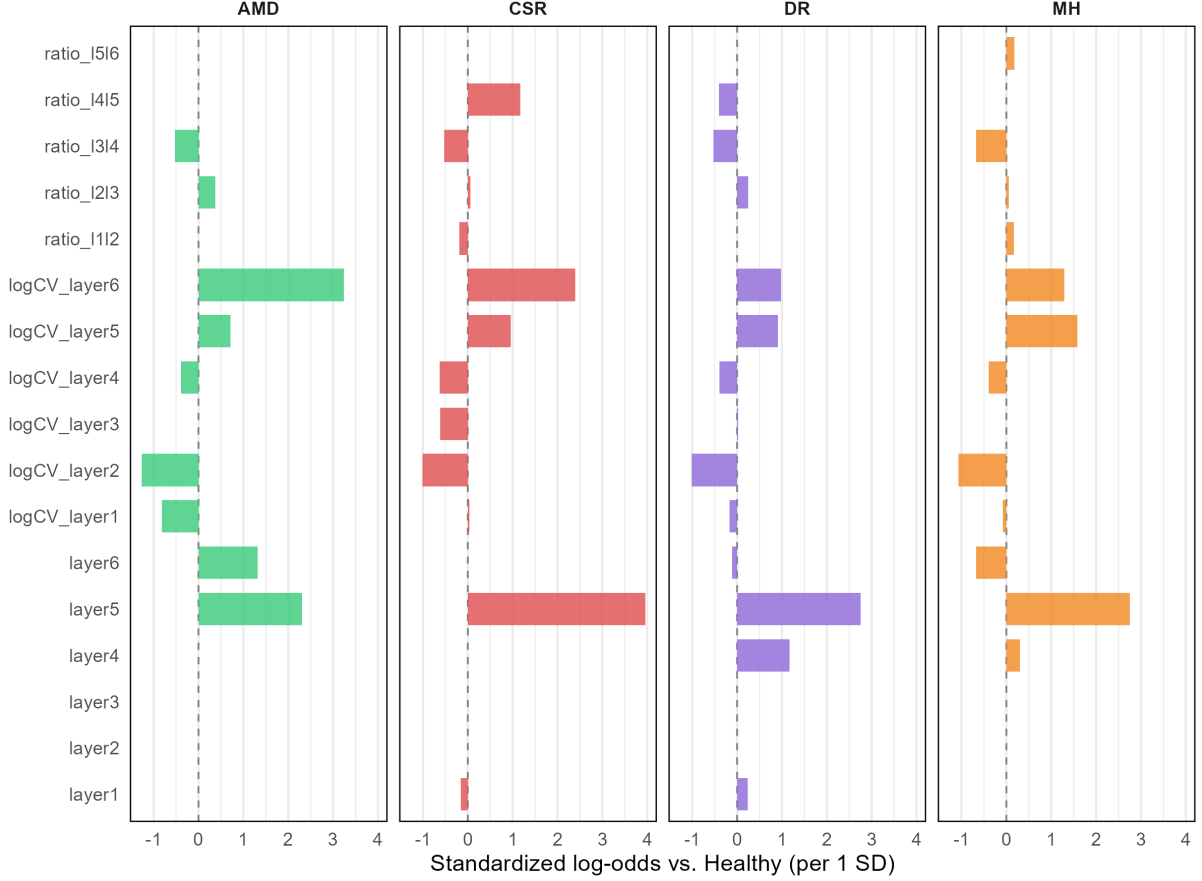


Figure 3.1: Standardized log-odds versus Healthy, $\tilde{\beta}_{kj}$ of Equation (3.5).

Interpretation

The lasso is aggressive: of the 17 predictors each class vector retains only seven to nine, and only the mean thicknesses of layers 2 and 3 are dropped from all five vectors at once.

Figure 3.1 concentrates the signal in the outer retina. Mean thickness of layer 5 carries the largest standardized log-odds in the figure (3.95 for CSR) and is positive for every pathology (2.75 for DR and MH, 2.31 for AMD), so a thickened layer 5 is a shared rather than a discriminating signature. What separates the classes is carried by the variability features and a few localised means. The relative spatial variability of layer 6 leads the AMD panel (3.24) with CSR behind it (2.40) and DR and MH much weaker (0.98 and 1.29); for that same layer the mean thickness moves independently, positive only for AMD (1.32), dropped for CSR and negative for DR and MH (-0.11 and -0.67), so roughness and overall thickness are separate pieces of information. Mean thickness of layer 4 is the most class-specific entry, appreciably positive for DR (1.17), small for MH (0.30) and dropped for AMD and CSR, and the ratio between layers 4 and 5 likewise splits CSR (1.17) from DR (-0.41) while absent elsewhere.

The inner retina contributes almost nothing through its mean thicknesses: layers 2 and 3 receive no weight in any class and layer 1 at most 0.24. What survives there is relative rather than absolute, and common to all pathologies rather than class-specific: the log coefficient of variation of layer 2 (-1.01 to -1.27) and the ratio between layers 3 and 4 (-0.53 to -0.68). The picture is thus one of discriminative structure concentrated in layers 4 to 6, with inner-retinal mean thicknesses largely uninformative after penalization.

3.3 Random forest

The second model examined in detail is the random forest, in its **ranger** implementation, fitted under the same nested cross-validation protocol of Section 3.1. The importance values discussed below are those of the configuration trained again on the `means_logCV_ratios` feature matrix. The forest is grown with 500 trees and the Gini splitting rule held fixed; as summarised in Table 3.1, the inner loop tunes two hyperparameters: `mtry`, the number of candidate predictors considered at each split, and `min.node.size`, the minimum size a node must have before it is split further.

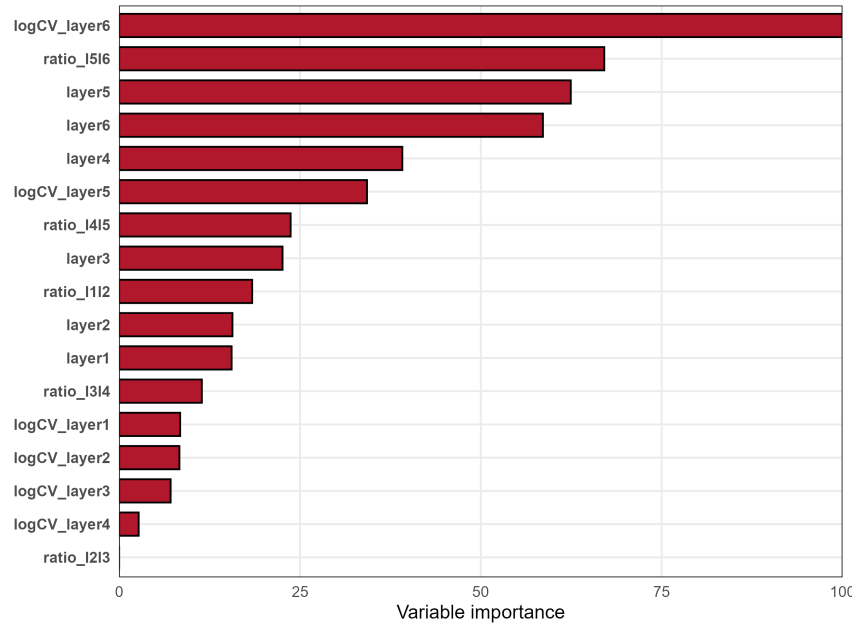


Figure 3.2: Impurity-based variable importance, rescaled to 0–100 relative to the maximum.

The impurity criterion accumulates the decrease in Gini impurity attributable to each variable over every split of every tree, a single non-negative quantity per feature. It is pooled across the five classes by construction: unlike the coefficient matrix of Section 3.2, it cannot say *which* pathology a feature helps to identify, only that the forest found it useful somewhere.

Interpretation

Figure 3.2 is led by the outer retina. The relative spatial variability of layer 6 (Table 1.1) attains the maximum score and sets the scale; a clear second band follows with the ratio between layers 5 and 6 (67) and the mean thicknesses of layers 5 (62) and 6 (59), then a lower tier of mean thickness of layer 4 (39) and the log coefficient of variation of layer 5 (34). Everything else scores at or below 24, the inner retina uniformly weak: mean thicknesses of layers 1 to 3 between 16 and 23, and the five lowest scores are the log coefficients of variation of layers 1 through 4 together with the ratio between layers 2 and 3.

This ordering is structurally consistent with Figure 3.1, which also puts its largest effects on the mean thickness of layer 5 and the relative variability of layer 6 and little weight on the inner-layer means. Two models from different families, a linear penalized classifier and a decorrelated tree ensemble, thus point to the same region, layers 4 to 6, and to the same split between absolute thickness and relative variability. The agreement cannot say which model is more discriminative, since the forest score is pooled across all five classes while the coefficients are class-specific; it does say that the outer retina, rather than the ganglion-cell and inner nuclear layers, is where these engineered features carry information about disease class.

Chapter 4

Functional data analysis

4.1 Smoothing

Each subject contributes six discretely sampled retinal-layer profiles (Table 1.1). To perform functional analysis, every profile is converted into a smooth function: $x_i^{(\ell)}(t)$ denotes the smoothed thickness profile of layer $\ell = 1, \dots, 6$ for subject i , evaluated at position t over the normalised domain $t \in [0, 1]$. For each subject and layer, the raw measurement vector is placed on an equally spaced grid over $[0, 1]$.

Specification

Smoothing is carried out with the `fda` package [6]. A cubic B-spline basis (order 4, piecewise cubic polynomials) is built once on $[0, 1]$ and shared by every subject and every layer; each profile is then fit to this basis. Writing $\{\varphi_j\}_{j=1}^J$ for the basis functions, a smoothed curve is represented as $x_i^{(\ell)}(t) = \sum_{j=1}^J c_{ij}^{(\ell)} \varphi_j(t)$, with coefficients obtained by penalized least squares,

$$\hat{c} = \arg \min_c \sum_j (y_j - x(t_j))^2 + \lambda \int_0^1 (D^2 x(t))^2 dt, \quad (4.1)$$

where $D^2 x$ is the second derivative of the fitted curve and $\lambda \geq 0$ is a roughness-penalty weight. The basis is fixed at $J = 40$ functions. This choice is meant only to place the basis comfortably above the complexity of the underlying signal, and basis sizes above 40 are largely absorbed by the roughness penalty in Equation (4.1) rather than translating into extra wiggleness of the fitted curves.

Smoothing is not governed by the basis size alone; the penalty weight λ in Equation (4.1) is chosen by generalized cross-validation (GCV) over the logarithmic grid $\lambda \in 10^{\{-4, -3.5, \dots, 4\}}$. For each candidate λ , `smooth.basis` returns a per-patient GCV score, and for each layer the λ that minimizes the *average* GCV score across all patients of that layer is kept. This gives one common $\lambda^{(\ell)}$ per layer, applied to every subject's profile in that layer, rather than a value optimized patient by patient. A shared λ is needed because the downstream analyses require every subject's curve to live in the same functional space. The six layers are smoothed independently of each other, six separate functional objects per subject, sharing only the basis and the domain, and each with its own layer-specific $\lambda^{(\ell)}$.

Figure 4.1 shows the outcome for two subjects, stacking the six fitted curves cumulatively so that the width of each band is that layer's thickness and the outer envelope is the total retinal thickness.

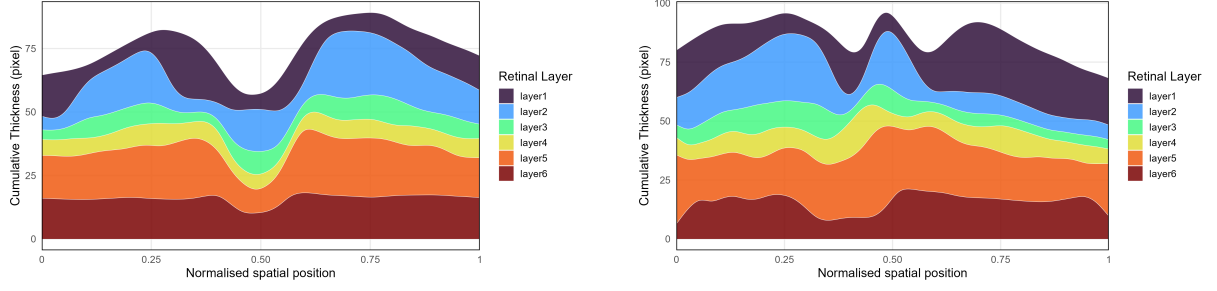


Figure 4.1: Healthy subject (patient 5, left) and AMD subject (patient 12, right).

Functional mean and variance

For a functional datum $x_i(t)$, $i = 1, \dots, N$, where N is the number of training subjects, the pointwise functional mean and variance are defined exactly as their scalar counterparts, evaluated separately at every t :

$$\bar{x}(t) = \frac{1}{N} \sum_{i=1}^N x_i(t), \quad s^2(t) = \frac{1}{N-1} \sum_{i=1}^N (x_i(t) - \bar{x}(t))^2. \quad (4.2)$$

Since the diagnostic groups are the object of interest, the code applies Equation (4.2) within each class k and each layer ℓ separately. Writing n_k for the number of training subjects in class k ,

$$\mu_k^{(\ell)}(t) = \frac{1}{n_k} \sum_{i \in k} x_i^{(\ell)}(t), \quad (\sigma_k^{(\ell)})^2(t) = \frac{1}{n_k - 1} \sum_{i \in k} (x_i^{(\ell)}(t) - \mu_k^{(\ell)}(t))^2. \quad (4.3)$$

Figures 4.2 and 4.3 show $\mu_k^{(\ell)}(t)$ and $(\sigma_k^{(\ell)})^2(t)$ for every layer and class.

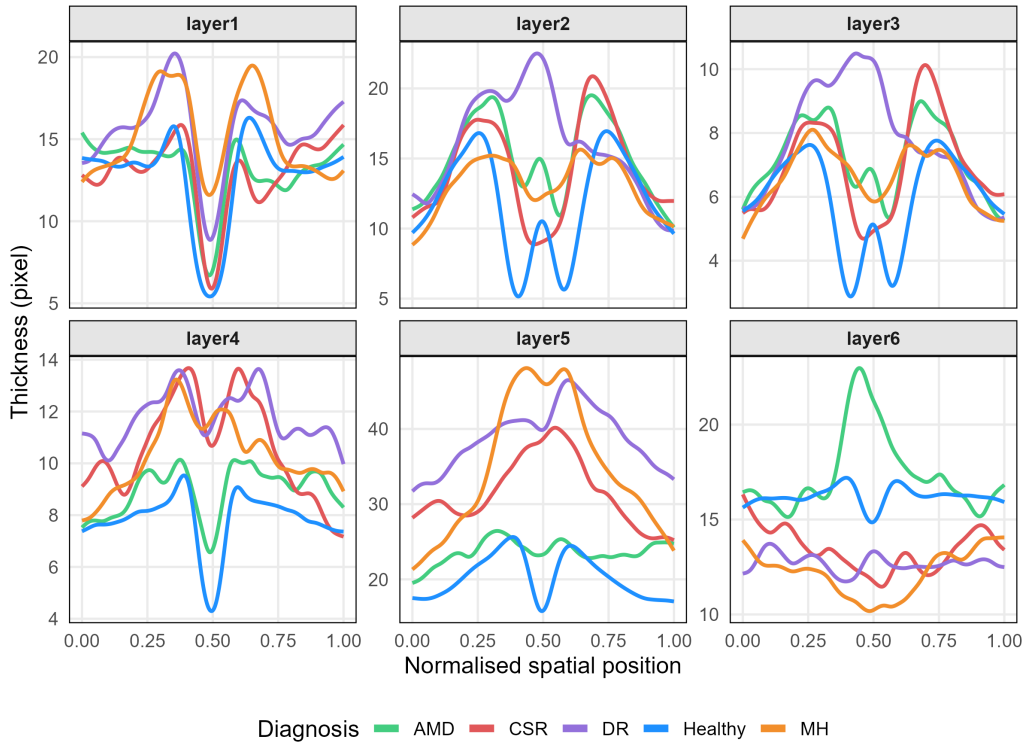


Figure 4.2: Mean functions $\mu_k^{(\ell)}(t)$.

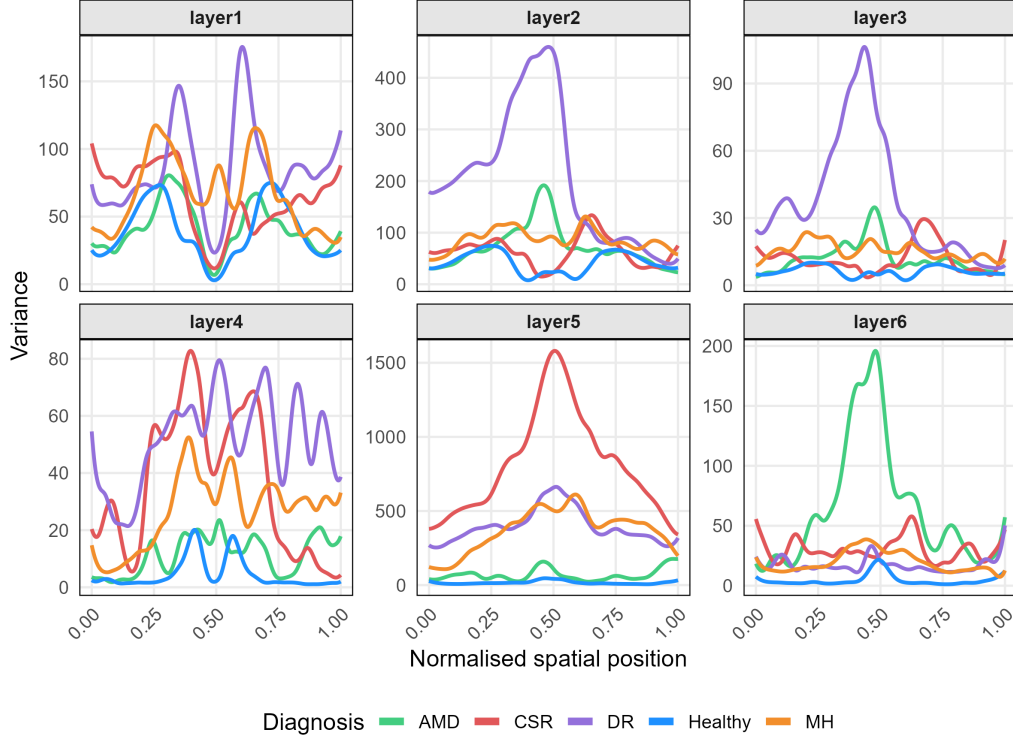


Figure 4.3: Variance functions $(\sigma_k^{(\ell)})^2(t)$.

Layer 5 (outer nuclear and photoreceptor inner region, Table 1.1) is by a wide margin the thickest layer in Figure 4.2: its mean values are roughly twice those of any other layer, and the Healthy curve runs below every disease class across the whole domain. Layer 5 also carries by far the largest pointwise variance in Figure 4.3, an order of magnitude above the other layers, and both the mean and the variance peak near the centre of the normalised axis, $t \approx 0.5$, the foveal region. The pattern is not uniform across layers: in layer 6, for instance, it is specifically the AMD class that develops a sharp, spatially localized peak in both mean thickness and variance near the centre of the domain, while the remaining four classes stay comparatively flat there. Taken together, the two figures indicate that thickness and cross-subject variability are concentrated in the same central, foveal region and largely in the same layer, layer 5, and that the disease classes differ from Healthy in spread as well as in level.

4.2 Class separability

It is useful to ask how much of the functional variability in the six smoothed layers is attributable to the disease class label, and where along the retina that class-related signal is concentrated.

Global effect size

The total functional dispersion in the product space $\mathcal{H} = [L^2(\mathcal{T})]^6$ is decomposed into a within-class and a between-class component,

$$W_k = \sum_{i \in k} \sum_{\ell=1}^6 \int_0^1 (x_i^{(\ell)}(t) - \mu_k^{(\ell)}(t))^2 dt, \quad B = \sum_{k=1}^5 n_k \sum_{\ell=1}^6 \int_0^1 (\mu_k^{(\ell)}(t) - \mu^{(\ell)}(t))^2 dt, \quad (4.4)$$

where $\mu_k^{(\ell)}(t)$ is the class- k mean function of layer ℓ (Equation (4.3)) and $\mu^{(\ell)}(t)$ is the grand

mean function of layer ℓ pooled over all training subjects. The functional analog of the ANOVA effect size is the ratio

$$\eta^2 = \frac{B}{B + \sum_{k=1}^5 W_k}, \quad (4.5)$$

the fraction of total functional variability explained by class membership.

For each smoothed layer, the $n \times n$ matrix of pairwise squared L^2 distances between subjects is computed. The six per-layer squared-distance matrices are summed to give the squared multivariate distance in the product space $\mathcal{H}$, and the square root of the result is passed to `vegan::adonis2` [5] as a distance object. `adonis2` runs a distance-based PERMANOVA of this distance matrix against the class label (9999 permutations, seed 2026). From its results η^2 can be extracted: for distances built from a sum of squared L^2 norms, the PERMANOVA sum-of-squares decomposition coincides with the trace decomposition in Equation (4.4). This makes η^2 a purely descriptive, trace-based measure: it aggregates across all six layers and all positions t into a single number, it does not model the covariance between layers.

The test rejects the null hypothesis of no difference between the class centroids very clearly: with 4 and 260 degrees of freedom the observed pseudo- F is 13.32, with $p = 10^{-4}$, the smallest value attainable with 9999 permutations. The associated effect size is $\eta^2 = 0.17$. The two numbers should be read together: the classes do differ, and the difference is far too large to be attributed to sampling variability, but class membership nevertheless accounts for only 17% of the total functional variability, the remaining 83% being within-class, a real but modest signal that limits how far any classifier built on these curves can go. The substantial within-class heterogeneity is consistent with the large within-class spread already visible in the pointwise variance functions of Section 4.1 (Figure 4.3).

Pointwise localization

A single aggregate η^2 does not say where along the retina the classes separate. To localize this, at each grid position t the six layer values $(x_i^{(1)}(t), \dots, x_i^{(6)}(t))$ are treated as one multivariate observation in $\mathbb{R}^6$, and a one-way MANOVA across the five classes is run at that position. The resulting Wilks' Lambda is

$$\Lambda(t) = \frac{\det \mathbf{W}(t)}{\det(\mathbf{W}(t) + \mathbf{B}(t))} \in (0, 1], \quad (4.6)$$

where $\mathbf{B}(t)$ and $\mathbf{W}(t)$ are the 6×6 between-class and within-class scatter matrices of the layer vector at position t . Unlike η^2 , $\Lambda(t)$ accounts for the covariance between layers at each position. This statistic is computed pointwise, independently at every position of the smoothing grid, by looping over it and fitting a `manova` at each position. Since small $\Lambda(t)$ indicates strong class separation, the plot represents $1 - \Lambda(t)$ rather than $\Lambda(t)$, so that large values correspond to strong separation.

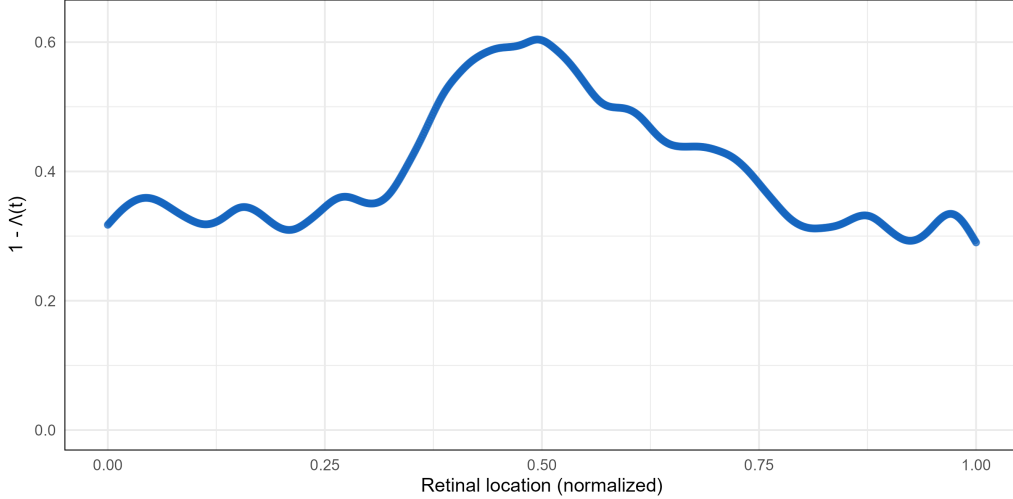


Figure 4.4: Pointwise class separation $1 - \Lambda(t)$.

Figure 4.4 shows a single pronounced peak centred at $t \approx 0.5$, the same central, foveal position already identified as where mean thickness and variance are largest in Section 4.1 (Figure 4.2, Figure 4.3). Away from this central region, $1 - \Lambda(t)$ settles at a markedly lower and comparatively flat level on both sides of the domain. Taken together with the modest η^2 above, this indicates that the class-related signal in these curves is not spread uniformly across the retina but concentrated in the central foveal region, even though class membership alone still leaves most of the total functional variability unexplained.

This pointwise display is used descriptively, to locate where along the retina class separation is strongest. Only the location and the relative shape of the curve are interpreted.

The central half of the scan, $t \in [0.25, 0.75]$, is taken as the foveal window used in the confirmatory analysis of Chapter 5. It is fixed a priori on anatomical grounds, covering the foveal and parafoveal region; Figure 4.4 confirms that it comfortably contains the region where $1 - \Lambda(t)$ is elevated.

4.3 Multivariate functional PCA

Multivariate functional PCA (MFPCA) is applied jointly to the six smoothed layer curves from Section 4.1, so that a single principal component is not a curve but a vector of six curves, one per layer, capturing a joint mode of variation shared across all six layers simultaneously. Writing $\mathbf{x}_i(t) = (x_i^{(1)}(t), \dots, x_i^{(6)}(t))$ for the six-layer profile of training subject i , MFPCA seeks the eigenfunctions of the covariance operator $\mathcal{C}$ of $\mathbf{x}_i$ on the product space $\mathcal{H} = [L^2(\mathcal{T})]^6$,

$$\mathcal{C}\boldsymbol{\psi}_k = \lambda_k \boldsymbol{\psi}_k, \quad \boldsymbol{\psi}_k(t) = (\psi_k^{(1)}(t), \dots, \psi_k^{(6)}(t)), \quad (4.7)$$

with each subject represented by scalar scores and reconstructed as

$$\xi_{ik} = \sum_{\ell=1}^6 \int_0^1 (x_i^{(\ell)}(t) - \mu^{(\ell)}(t)) \psi_k^{(\ell)}(t) dt, \quad \mathbf{x}_i(t) \approx \boldsymbol{\mu}(t) + \sum_{k=1}^M \xi_{ik} \boldsymbol{\psi}_k(t), \quad (4.8)$$

where $\mu^{(\ell)}(t)$ is the grand mean function of layer ℓ (Section 4.2) and M is the number of retained components. The same score ξ_{ik} multiplies all six components of $\boldsymbol{\psi}_k$, so the six layers are not treated independently: each component encodes one joint spatial pattern across all layers, not six separate per-layer patterns.

MFPCA is fit through the `MFPCA` package [2]. Because the eigenfunctions of $\mathcal{C}$ form in principle an infinite system, a percentage of variance explained is only meaningful relative to a

finite truncation. The number of retained multivariate components is fixed at $M = 10$. The proportion of variance attributed to each component is then computed relative to the retained set, which is the share each component explains among the components kept, not a share of the infinite total variance.

Leading eigenfunctions

Figure 4.5 shows the first six multivariate eigenfunctions ψ_k , the leading modes of joint patient-to-patient variation among the ten retained components, which together account for approximately 90% of the variance of the retained set. Each panel is one component ψ_k ; within a panel, the six coloured curves are its layer-specific components $\psi_k^{(\ell)}(t)$. The decay is steep: ψ_1 alone carries about 52% and ψ_2 a further 12%, so the joint variation of the six layers is concentrated in very few directions.

The components are not all of the same kind. Layer 5 dominates ψ_1 and ψ_3 by a wide margin, with the other five layers close to zero: the first is a single-signed bulge peaking at mid-domain, the third changes sign around $t \approx 0.6$, so between them they describe the amplitude and the left-right balance of layer 5 and little else. The remaining components are mixed rather than layer-5 modes with small contamination. In ψ_2 the largest excursion belongs to layer 2, opposed to layer 5 over the first half of the domain; ψ_4 and ψ_5 combine layer 5 with layers 1 and 2 at comparable amplitude; and by ψ_6 layers 1 and 2 lead outright, in a curve that oscillates several times across the domain. Higher-order components thus capture finer, more localized structure, and progressively relocate it from the outer retina towards the inner layers.

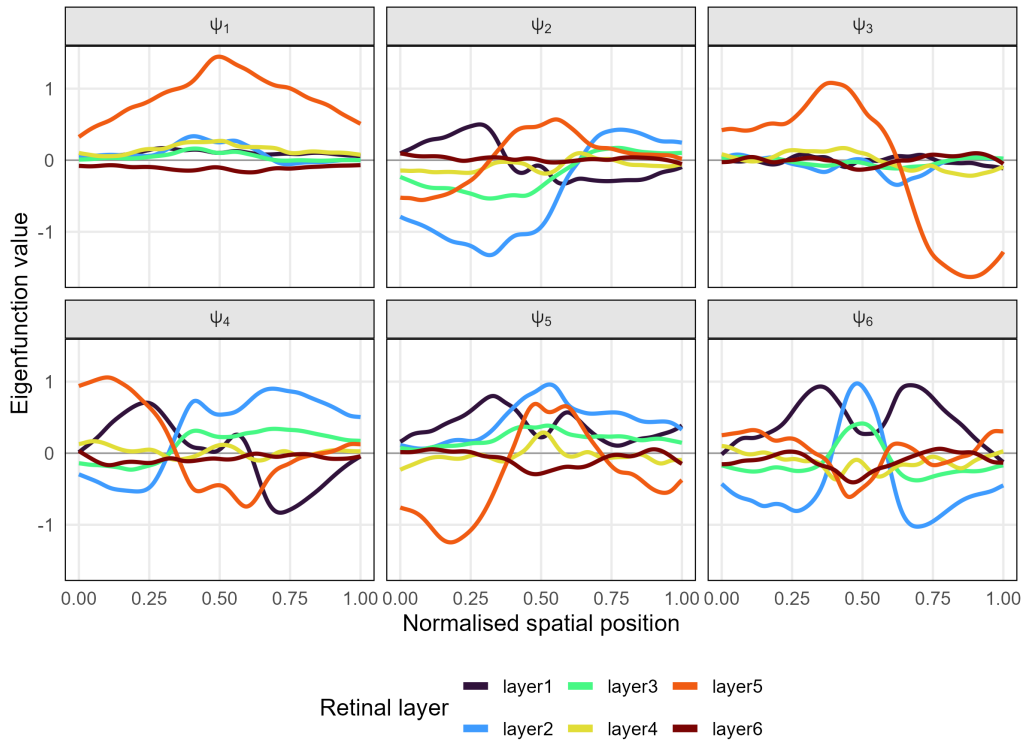


Figure 4.5: Multivariate eigenfunctions $\psi_k^{(\ell)}(t)$.

Perturbation plots

Raw eigenfunctions are hard to interpret in isolation, since a curve such as $\psi_k^{(\ell)}(t)$ has no units and no baseline to be read against. Each component is therefore visualized structurally, by

perturbing the six-layer mean function along it by a fixed multiple of the component's own standard deviation [7],

$$\mu^{(\ell)}(t) \pm c\sqrt{\lambda_k}\psi_k^{(\ell)}(t), \quad c = 2, \quad (4.9)$$

where λ_k is the eigenvalue of Equation (4.7), that is the variance of the scores ξ_{ik} on component k , so that $\sqrt{\lambda_k}$ is their standard deviation. The two perturbed profiles are what a subject two score standard deviations above and below the mean along component k alone would look like, with every other score held at zero. Scaling each component by its own $\sqrt{\lambda_k}$ puts the rows of the figure on a comparable footing: the gap between the two dashed curves is the amount of variation the component actually carries. By construction this display says nothing about the diagnostic classes, it just describes the shape of each mode of variation.

Figure 4.6 shows these perturbed curves for the first 4 components across all six layers.

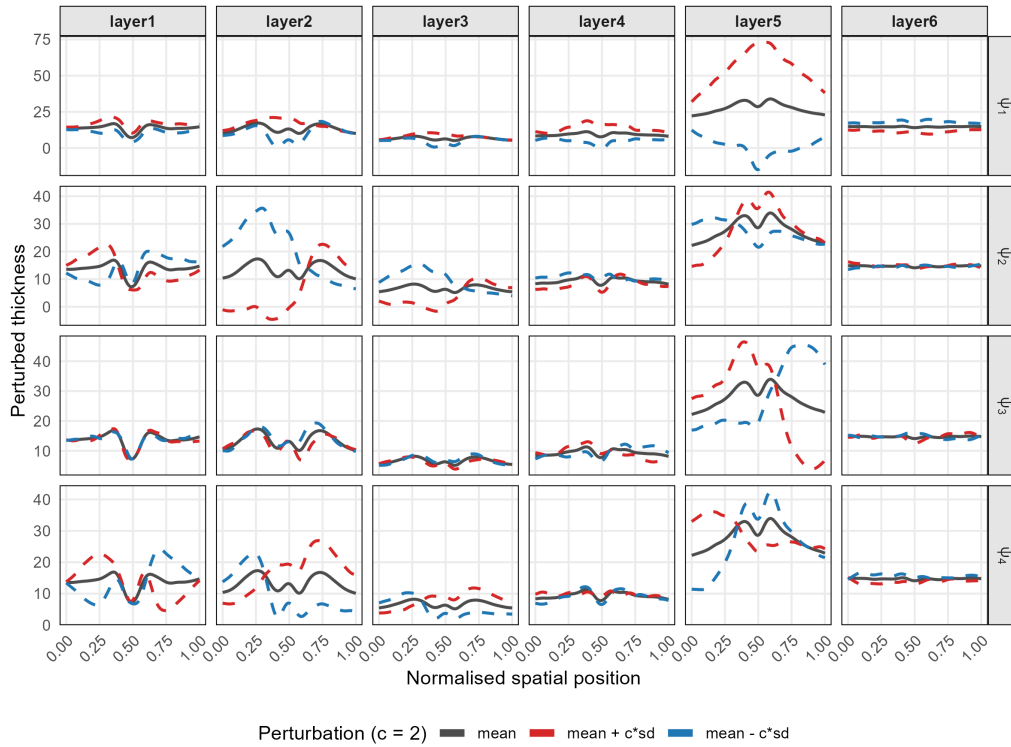


Figure 4.6: Rows are components $\psi_1, \dots, \psi_4$, columns the six retinal layers.

Three of the four components move layer 5 several times more than any other layer, and over the central portion of the domain: the same foveal region identified as thickest and most variable in Section 4.1 and as the site of strongest separation in Section 4.2. The remaining layers are nearly flat, except layer 2 in ψ_2 , the one component where layer 5 is not dominant.

The shapes differ. In ψ_1 the two perturbed curves stay on opposite sides of the mean across the whole domain, lifting the central peak from roughly 35 to above 70 in one direction and flattening it in the other, so ψ_1 measures how pronounced the foveal bulge is. In ψ_3 and ψ_4 the curves cross near mid-domain and exchange sides, so thickening on one side of the fovea accompanies thinning on the other: these encode the position and asymmetry of the bulge rather than its size, as does ψ_2 in layer 2. The lower ψ_1 curve falls below zero near the centre, which only reflects that Equation (4.9) extrapolates a single mode at two standard deviations rather than describing any observed patient; the sign and shape of the excursion, not its height, is what the panel supports.

Chapter 5

Validation and conclusions

5.1 Confirming spatial localization

The signal identified so far is concentrated in the outer retina and, spatially, around the fovea, but every result establishing that is associative. A large standardized log-odds or importance score (Sections 3.2, 3.3) says a feature is weighted heavily by a fitted model; a sharp peak in $1 - \Lambda(t)$ (Section 4.2) or a perturbation concentrated in layer 5 over the central domain (Section 4.3) says where the data vary most between classes. This section subjects the localization claim to a more direct, confirmatory test: if the class signal really is concentrated in layers 5 and 6 over an anatomically pre-specified central window, a classifier restricted to that subset of the raw measurements, refit from scratch, should recover approximately the performance of the unrestricted baseline on unseen patients. A large drop would instead show that the discarded periphery and inner layers were carrying discriminative information of their own.

Restricted configurations

Four versions of `means_logCV_ratios`, the feature matrix retained by both models in Sections 3.2 and 3.3, of increasing restriction and beginning from the unrestricted baseline, are compared:

- **full**: the unrestricted 17-feature baseline (6 means, 6 log-CVs, 5 adjacent ratios), all 6 layers over the whole domain.
- **central**: the same 17 features, recomputed after restricting every raw profile to the central window $t \in [0.25, 0.75]$, pre-specified on anatomical grounds as the foveal and parafoveal region and consistent with the elevated region of Figure 4.4, not tuned on any performance metric; the standard deviation, and hence the log coefficient of variation, is recomputed over the retained positions only.
- **layers56**: the full domain, but keeping only the marginal features that name layer 5 or layer 6 (2 means, 2 log-CVs, and the two ratios that touch either layer, `ratio_1415` and `ratio_1516`), together with the patient-specific correlation between the two layers (L_{5-6} , Section 2), 7 features in total, all of them referring to the layers implicated in Sections 3.2 and 3.3.¹
- **central_layers56**: the intersection of the two restrictions, the same 7 features recomputed entirely on the central window, so that no retained feature refers to the periphery.

Sections 3.2 and 3.3 place the discriminative structure in layers 4 to 6, so restricting to layers 5 and 6 alone is deliberately narrower than what those sections establish: layers 5 and 6 are

¹`ratio_1415` draws on the layer-4 mean, so the effective share of raw measurements is slightly larger than the nominal 33.3%. The same applies to `central_layers56`, whose nominal share is 16.6%.

retained because they carry the strongest and most class-specific effects in both models, and because confining the restriction to them makes the anatomical claim under test a sharper one. Layer 4 is not excluded outright, since it still enters indirectly through `ratio_1415`.

Each restriction is fixed before any test-set evaluation, on the anatomical and analytical grounds already established earlier in the report: the central window on anatomy, the layer selection on the training-set findings of Sections 3.2 and 3.3. No restriction is informed by test-set performance. Although this yields eight test-set evaluations (four configurations times two models), all four restrictions were pre-specified on anatomical and training-set grounds before any use of the test set, so the test set is not spent on configuration selection.

Results

For every configuration, the penalized logistic regression and the random forest are refitted. Both go through the identical nested cross-validation protocol of Section 3.1, with the same ten outer folds, the same seed, and the same hyperparameter grids of Table 3.1, so every configuration receives its own tuned hyperparameters. Only after this refitting is complete is the resulting final model evaluated once on the 86 patients of the held-out test set.

Dataset	Feat.	Domain	Macro-F1	Bal. acc.
full	17	100%	0.671	0.803
central	17	49.9%	0.702	0.820
layers56	7	33.3%	0.624	0.774
central_layers56	7	16.6%	0.602	0.754

Table 5.1: Penalized logistic regression. “Domain” is the share of raw measurement cells the configuration still sees.

Dataset	Feat.	Domain	Macro-F1	Bal. acc.
full	17	100%	0.703	0.816
central	17	49.9%	0.621	0.766
layers56	7	33.3%	0.657	0.794
central_layers56	7	16.6%	0.677	0.794

Table 5.2: Random forest, same configurations.

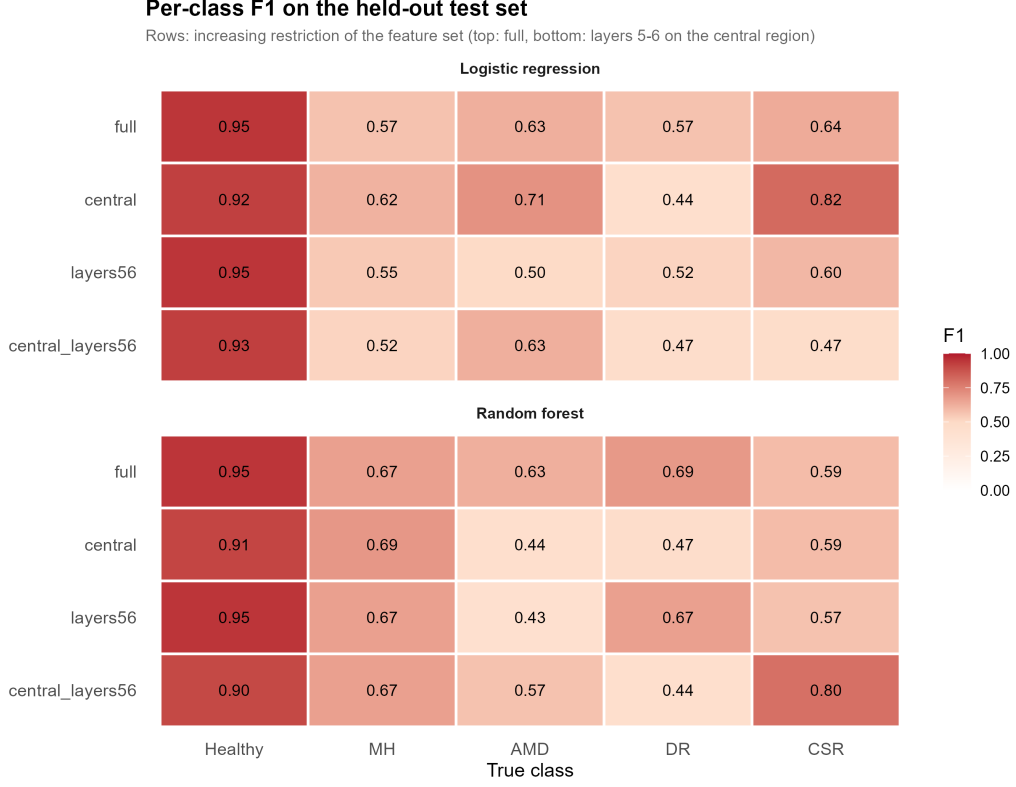


Figure 5.1: Per-class test-set F1. Rows are configurations in order of increasing restriction, panels the two models.

No configuration collapses: none of the eight macro-F1 values of Tables 5.1 and 5.2 falls below 0.60, not even **central_layers56**, which retains about a sixth of the raw measurements. The largest deviation from **full** (-0.082) is smaller than the across-fold standard deviations of Table 3.2 (0.059 to 0.168), so the ordering within each table is indicative rather than a powered contrast. The claim the experiment does support is the coarse one: discarding the periphery and the marginal features of four of the six layers, with layer 4 surviving only indirectly through **ratio_1415**, does not empty the classifier of discriminative information.

The two models disagree on which restriction costs the most. The layer restriction costs both almost the same (-0.047 macro-F1), but the central window alone is a small gain for the logistic regression ($+0.031$) and the largest loss for the random forest (-0.082). Figure 5.1 attributes these to single classes: the logistic regression’s gain is CSR ($0.64 \rightarrow 0.82$, consistent with subretinal fluid collecting under the fovea), the random forest’s loss is AMD ($0.63 \rightarrow 0.44$) and DR ($0.69 \rightarrow 0.47$). Healthy is recovered almost perfectly throughout (0.90 to 0.95) and MH shows no consistent trend (0.52 to 0.69), while DR is the one class that declines under every restriction in both models, as would be expected of a diffuse vascular condition rather than a focal macular one.

The two **full** rows also invert the ranking on which the models were selected. Under the nested cross-validation of Section 3.1 the logistic regression led the random forest (0.6528 against 0.6028, Table 3.2); on the test set the order is reversed (0.671 against 0.703). Both movements lie within the across-fold standard deviations of those estimates (0.1025 and 0.1116), and a macro- F_1 averaged over five classes of which the smallest contributes 9 patients has a standard error of the same order, so this is sampling variability, not evidence that the random forest was the stronger model all along. This is precisely why selection was carried out on the cross-validation estimates: the pooled out-of-fold estimate averages over ten folds and the whole training set, whereas a single evaluation on 86 patients is too noisy to arbitrate a gap of this size. The cost is

that the selected model is not the test-set winner; the benefit is that the test-set figures remain an estimate of generalization rather than a spent selection criterion.

Two caveats bound how far this can be pushed. First, a restriction can only remove information, yet it may still measure better: dropping predictors that are mostly noise for a given model reduces the variance of the fit, and that reduction can outweigh the information lost. Where a restriction does appear better, as the logistic regression is on the central window (+0.031), that is the mechanism to read into it, not added signal. Second, the test set holds 86 patients, of whom only 9 are AMD and 10 CSR (Table 1.2): a single patient changing side moves a per-class F1 by several hundredths, the order of magnitude of the differences discussed above.

The localization claim therefore survives, but not as a single shared signature: the fovea and layers 5–6 carry most of what separates AMD and CSR from Healthy, and appreciably less of what identifies DR.

5.2 Conclusions

This report asked where in the retina disease leaves a detectable structural signal, and what that signal corresponds to anatomically. Three stages of analysis, differing in the representation they impose on the same thickness profiles, converge on one answer: the signal lives in the outer retina, layers 4 to 6, over the central foveal portion of the scan, and the inner layers contribute almost nothing through their mean thicknesses. That the penalized linear model, the tree ensemble and the functional decomposition agree on this is the substance of the claim, since none of the three is committed to the others' assumptions.

Two distinctions matter for how the finding should be read. The first is between a marker of disease and a marker of *which* disease: layer 5 thickens in all four pathologies, and what separates the classes from one another is carried instead by relative spatial variability, chiefly in layer 6, and by a few localized means. Absolute thickness and its variability are separate pieces of information, and only the second discriminates. The second distinction is between where the data vary most and where the discriminative information sits. The confirmatory experiment of Section 5.1 separates them: classifiers refitted from scratch on the central half of the scan, and on layers 5 and 6 alone, recover macro-F1 comparable to the unrestricted baseline while retaining about a sixth of the raw measurements. Localization is therefore confirmed rather than merely suggested, though not as a single signature shared by all four classes: the restricted models lose appreciably more of what identifies DR, than of what identifies the focal macular ones.

The ceiling on all of this is that class membership explains only 17% of the total functional variability. The remaining 83% is within-class, and it is not noise: the classes that shift the mean most also inflate the pointwise variance most, so disease produces both a change in thickness and a more variable presentation. A biomarker can be real and anatomically coherent, as this one appears to be, and still be insufficient on its own to classify an individual patient.

Two limits bound the conclusions. The test set holds 86 patients, nine of them AMD and ten CSR, so per-class figures move by several hundredths when a single patient changes side. And the analysis is a discovery exercise throughout: the layers, the region and the features were identified on these data, and the confirmatory experiment tests whether the localization survives restriction, not whether it reproduces elsewhere. The natural continuation is validation on an independently acquired cohort, together with a scalar-on-function treatment that would estimate a weight function along the retina directly, in place of the summaries through which the spatial structure has been approached here.

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